Serious Games for Social Innovation

Reviewing Their Impact on Citizen Participation and Spatial Justice

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Abstract – This study explores the possibility of applying serious games to increase citizen participation and foster spatial justice in urban decision-making processes. Drawing on theories of spatial justice, social innovation, and participatory spatial planning, it investigates how game-based approaches can smooth dialogue between citizens, experts, and other stakeholders. The focus is to bridge the gap between citizens' needs and demands and planners' goals and actions. A serious game, conducted with stakeholders, served as an experimental platform to visualize, simulate, and negotiate spatial interventions in an engaging and interactive atmosphere. The findings indicate that this approach not only strengthens citizens engagement but also fosters mutual understanding among actors with different interests. By transforming abstract planning concepts into accessible and collaborative processes, participatory games empower marginalized voices and support the pursuit of more just and sustainable urban strategies. This study concludes that gamification can be an effective and scalable tool for advancing spatial justice in socially diverse urban contexts.

Introduction

Spatial planners play a crucial role in creating social changes through their decisions. Persistent environmental and socio-economic challenges in urban areas have prompted a reevaluation of innovation's role within sustainable urban development. Sustainable development is understood as a paradigm fostering activities that respect social limits while promoting environmental and economic progress. This approach embodies social innovation a transformative force capable of reshaping human life and permeating all sectors of society. Social innovation contributes significantly to sustainability by replacing outdated social practices with new ones that foster a sustainable society (Nyseth & Hamdouch, 2019).

Bosse (2025) argues that citizen participation faces multiple barriers, as decision-making is often dominated by governments, while other actors such as citizens, parliaments, civil society, and local authorities have limited power. Participatory tools are fragmented, difficult to access, and often undermined by low voter turnout, political apathy, and restricted civic space. Reforms tend to be top-down, weakening public ownership and accountability. Referendums, though a form of direct democracy, are frequently misused and fail to encourage informed debate. Similarly, Tan (2014) notes that large-scale urban projects prioritize speed over inclusive engagement, leading to unaccounted social, environmental, and economic costs. In contrast, sustainable urban development depends on transparent, democratic processes where stakeholders define real needs. This approach aligns the goals and actions of experts with citizens' needs and demands, thereby helping to avoid greater long-term costs.

In response to these challenges, new engagement methods such as game-based approaches have emerged. These interactive systems leverage actors' interactions to find and test urban solutions, offering normal citizens transparent, engaging, and easy-to-understand way to participate in spatial planning (Dodig & Groat, 2019a; Ferri et al., 2018; Poplin, 2011; Tan, 2017).

According to Marome, Natakun, and Archer (2021), serious games facilitate interactive storytelling by providing players with simulations that challenge them to think critically, assess issues from fresh perspectives, and reconsider their roles (Cruz-Cunha, 2012; Lean, Moizer, Towler, & Abbey, 2006). Participatory games, characterized by their experimental, creative, and democratic nature, engage actors in urban change by allowing minor interventions and scenario testing within accessible game simulations. These methods translate complex urban relationships into simple rules, fostering negotiation and collaboration despite conflicting interests (Thompson, 2019). Ultimately, this participatory, game-based approach supports self-organizing urban development and offers promising strategies for bottom-up planning. This study examines how game-based participatory methods support co-creation and inclusive decision-making toward achieving spatial justice.

Citizen Participation and Spatial Justice

The concept of spatial justice has roots in Aristotle's political philosophy, emphasizing a polity where citizens respect each other's rights and governance is by the best-qualified with consent (Kenny & Amadio, 2025). It evolved through liberal democracy and urban crises of the 1960s, notably influenced by Lefebvre's "*Right to the City*" and Foucault's spatial theories. David Harvey further developed the idea of "*territorial justice*" in Social Justice and the City (1973), which shaped ongoing debates on justice and democracy. Soja (2010) later framed spatial justice as a critical spatial perspective complementing social and economic justice, aiming to foster inclusive democratic politics and grassroots activism. Justice supports social sustainability because as Rawls claimed (1971):

> "Justice is the first virtue of social institutions, as truth is of systems of thought. A theory however elegant and economical must be rejected or revised if it is untrue; likewise, laws and institutions no matter how efficient and well-arranged must be reformed or abolished if they are unjust."[1]

As Rocco (2023) mentioned, when examining how social justice manifests within urban environments, we prefer the term '*Spatial Justice*' because it explicitly highlights the spatial dimensions involved in fairly distributing the burdens and benefits of human coexistence in cities and communities. This perspective emphasizes recognition of diverse ways of living, different relationships with the environment, and the distinctive processes by which spatial relationships are governed.

Spatial justice addresses the geographic dimensions of fairness and inequality, highlighting the uneven allocation of resources and opportunities across places. It exposes systemic causes of spatial injustice such as class, race, and gender discrimination, as well as exclusionary political and economic practices like redlining and gerrymandering (Soja, 2010). Although perfect spatial justice is unattainable, strategic interventions can mitigate deep-seated inequalities. Spatial justice is integral to

1 Rawls, J. (1971). A theory of Justice. Cambridge, Mass.: Harvard University Press.

urban sustainability, linking social, economic, and environmental dimensions to enhance livability and intergenerational equity (Rocco, 2023).

At TU Delft Department of Urbanism, conceptualizes spatial justice in three crucial dimensions: distributive justice (the spatial distribution of the burdens and benefits of our human association in cities and communities), procedural justice (the justice in the procedures and governance of the built environment), and recognition justice (the acknowledgment, validation, and respect for individual and collective identities, experiences, and cultural expressions that determine historical and ongoing marginalization, discrimination, and misrepresentation of certain groups in society). Together, these three dimensions form a comprehensive framework for spatial justice, which is vital for sustainable democratic governance and empowering citizens to influence the equitable shaping of urban environments. As demonstrated in Fig. 1, the three dimensions of spatial justice are interdependent and mutually reinforce one another (Rocco, 2023).

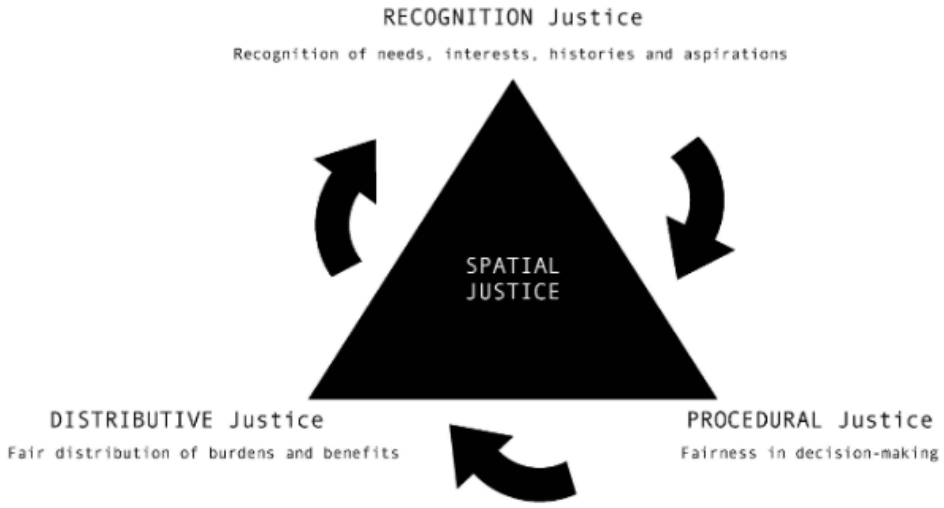


Figure 1: *The three dimensions of spatial. (Source: Rocco, 2023).*

The given diagram created by author summarizes the connections between spatial planning, sustainable development, spatial justice, social innovation, citizen participation, and gaming. According to the three dimensions of spatial justice citizen participation becomes important, particularly when considering the emphasis of *procedural justice* on fairness in decision-making.

Leventhal et al. (1980) provided a list of six criteria that are used to decide whether a procedure is just:

- The process should treat all parties consistently.
- Be free from bias.
- Use accurate information in rendering decisions.
- Take into account the views of all (something like voice).
- Be correctable in the event of an error, and remain consistent with prevailing ethical norms.

These six criteria have remained influential, although some additional rules might also be important in certain settings (Cropanzano & Molina, 2015).

As shown in figure 2 (Rocco, 2023), the triangle of governance represents the interactions among civil society, the public sector, and the private sector in shaping cities. Procedural justice focuses on how these actors plan and manage urban spaces through both formal institutions and informal norms. Transparent, participatory plan-

ning that includes diverse voices helps ensure a wider range of needs and aspirations are addressed (Rocco, 2023).

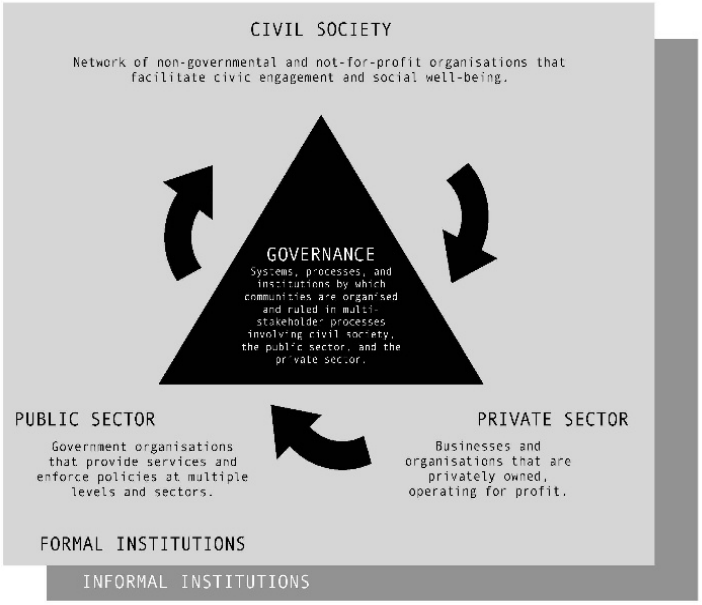


Figure 2: *The triangle of governance. (Source: Rocco, 2023).*

Serious Games

Games, as structured forms of play, represent a fundamental and multifaceted human activity with deep cultural, educational, and social implications. As Johan Huizinga (1938) mentioned:

> "Play is older than culture, for culture, however inadequately defined, always presupposes human society, and animals have not waited for man to teach them their playing."[2]

This foundational perspective underlines the significance of play and games as more than mere entertainment they are pivotal to cultural development and human flourishing.

Games, defined by clear rules and goals, enable interactive and collaborative engagement. Huizinga (1938) and later scholars highlight games as early learning mechanisms, with Richard D. Duke (1974) describing games as the "*language of the future*," underlining their potential in education and problem-solving. While Ludwig Wittgenstein famously claimed games are indefinable, Suits (2005) countered by defining games as voluntary attempts to overcome unnecessary obstacles, situating play at the core of ideal human existence. Juul (2013) expanded on the nature of games, focusing on the unique player experience of failure and learning as a source of engagement and satisfaction.

SERIOUS GAMES AS TOOLS FOR PARTICIPATORY URBAN PLANNING

The concept of serious games, introduced by Abt (1987), extends beyond entertainment into purposeful simulations used in areas such as defense, education, healthcare, and increasingly, urban planning. These games balance educational value with

2 Homo Ludens : a study of the play element in culture by Huizinga, Johan, 1872-1945

engaging experiences, making them effective tools for participation and discourse (Marome, Natakun, & Archer, 2021).

Game Theory provides a useful foundation for understanding how serious games can foster cooperation in participatory contexts. It models interactions under the assumption that stakeholders share information and work toward equilibrium, helping to structure negotiations and decision-making (Procol, 2022). In urban processes, cooperation means aligning individual interests toward collective goals—enabled by trust, reciprocity, and mutual benefit (Pidathala, 2023).

Serious games have garnered increasing attention in urban design and planning due to their ability to support participatory processes. They enable citizens to explore their living environments, engage in future speculation, and contribute to shaping development outcomes (Bogost, 2010; Morschheuser et al., 2017; Tilvawala et al., 2019). Early planning applications emerged in the 1960s with playable simulations (Duke, 2011). With the rise of collaborative planning (Healey, 1992; Innes & Booher, 2018), these games evolved toward more player-centered interaction, emphasizing negotiation and managing complexity in urban contexts (Tan, 2016).

Serious games support civic engagement by providing interactive platforms for co-creation between citizens and planners. This enhances ownership of urban infrastructure and strengthens long-term planning commitment (Duckworth, as cited in Cranenburgh, 2019). Key design principles include mechanics, aesthetics, technology, storytelling, and briefing to guide dynamics and expectations (Schell, 2008; Fabricatore, 2007; Björk, 2009). Boller and Kapp (2017) further highlight essential game elements such as goals, rules, feedback, challenges, interaction, and emotional engagement—especially relevant to participatory planning.

One prominent approach is *city gaming*, exemplified by the Generative City Game. These games allow players to assume stakeholder roles and make collaborative decisions about urban change. City gaming functions as a multi-agent platform, a form of "free-form gaming" that invites real-life unpredictability into the gameplay (Tan, 2014). Unlike rigid simulations, free-form gaming reflects the complexity of urban life through flexible, player-driven interactions.

By translating urban dynamics into simplified game rules, city gaming activates collective intelligence among diverse stakeholders. It fosters community through playful engagement, encouraging shared visions and accountability in implementing urban strategies (Tan, 2010b).

Tan (2018) notes that Game Theory can also inform procurement and negotiation processes by introducing rule-based structures that enhance transparency and competition. However, participatory urban planning often requires adaptive rules that reflect lived experiences rather than fixed assumptions. These evolving frameworks accommodate context-specific decision-making (Tan, 2017).

To support such adaptive engagement, Kapp (2012) identifies four types of game rules:

- Operational goals: define how to win or succeed in the game.
- Constitutional rules: maintain smooth and fair gameplay.
- Implicit behavioral rules: guide acceptable player behavior.
- Instructional rules: support learning during the game process.

SERIOUS GAME AS A SOCIAL INNOVATION TOOL

Public participation has benefited from innovative engagement tools that facilitate negotiation among diverse stakeholders with varying planning perspectives (Sousa, Antunes, Pinto, & Zagalo, 2022). Graeber's (2015) work introduces the concepts of play and games to offer a fresh perspective on transformative social innovation, emphasizing its foundation in spontaneous, creative play. He also explores rules and bureaucracy alongside the role of play and creativity in institutional structures. Additionally, he argues that rigid bureaucratic systems often stifle innovation and creativity, highlighting the relevance of "play" and "games" in fostering social innovation and institutional change.

The global population faces numerous social challenges daily, which has led to increased focus on innovative solutions. Social innovation, a key concept, emphasizes harnessing creativity to develop effective solutions while efficiently using limited resources to address social problems (Avelino et al., 2019). Scholars describe social innovation as a collaborative response to human needs and shared values that emerge from interactions among diverse actors (Correia et al., 2018). This field generates solutions driven by various agents organized under different legal frameworks and is increasingly recognized by managers as a strategic dimension in organizational planning (Andrade et al., 2021). As Wirman (2018) notes, social innovation encompasses a broad range of practices aimed at improving society, especially for marginalized groups. Osburg (2013) captures this normative aspect by stating social innovation "usually implies that something positive is created for society."

Another approach addressing social challenges is serious games, which serve purposes beyond entertainment. Olejniczak et al. (2020) identify four key characteristics enhancing their effectiveness: use of a universal language, flexibility to navigate uncertainties and complexities, facilitation of learning, and enabling timely data collection. These features position serious games as powerful tools for tackling social issues.

Rocha (2018) highlights how games enable players to explore imaginary spaces and manipulate symbolic systems, fostering critical thinking and opinion formation. Games are cultural artifacts capable of persuasion and meaning-making, with the potential to drive long-term social change. A bibliometric analysis by Van Der Have and Rubalcaba (2016) shows social innovation is often conceptualized as a creative strategy impacting social relations and structures, addressing needs while promoting social equity and collective well-being (Taylor, 1970; Gentil et al., 2019).

Comparing these fields reveals significant overlaps: both serious games and social innovation address key social issues through systematic and creative strategies. They offer solutions that meet human needs and promote positive change in social systems, behaviors, education, natural resource management, social responsibility, health, media, citizen inclusion, empowerment, and other socially relevant areas (Schrier, 2016; Wilkinson, 2016; Van Der Have & Rubalcaba, 2016; Mayer et al., 2016).

In urban development, planning-based participation methods typically focus on social and political dynamics through agency negotiations, while design-based methods emphasize structured, iterative approaches with clear visual outcomes. Both approaches have limitations: planning often struggles to address the evolving physical environment, and design lacks effective modeling of decision-making processes. Gaming emerges as an innovative social approach bridging these gaps by facilitating knowledge creation and negotiation, integrating diverse stakeholders, and enhancing community-informed policies and design solutions (Tan, 2017).

Finally, viewing serious games as a form of social design broadens their relevance beyond youth engagement. This perspective highlights their potential to engage diverse age groups and foster intergenerational dialogue and participation. Recognizing this broader capacity enables serious games to address social issues inclusively across demographics, promoting solutions resonant with a wide audience.

Discussion

Serious games are becoming popular tools in urban planning, but their use often doesn't lead to real, creative participation from the public. Instead of people actively shaping plans together, public input is usually limited to feedback after decisions have been made (Indraprastha & Shinozaki, 2009; O'Coil & Doughty, 2004). This shows a key problem: even though serious games are interactive, they don't always work as true platforms for democratic, collaborative planning.

Recent studies highlight the need for planners to create tools that not only engage citizens but also encourage learning and shared decision-making. At the same time, planners face limited resources and some doubt from professionals in the field (Ampatzidou et al., 2018). This means it's important to find a balance between practical limits and the hopeful goals of participation.

There is a tension between free-form play and structured games, as Thompson (2019) explains. Real change needs the freedom and creativity of play, but also rules and order to make innovations last. Graeber (2015) adds that while games have clear rules and goals, play allows surprises and new ideas to emerge within those rules.

Serious games bring together different aspects of city planning, such as government actions, market forces, people, and physical features like roads and buildings. Because of this, they can help with complex decisions and policy-making. Mayer's research shows these games encourage experts from different fields to work together by creating interactive discussions at many levels (Raghothama & Meijer, 2015). According to Raghothama and Meijer (2015), games that act like Planning Support Systems (PSS) can support urban decision-making by being collaborative, involving multiple players, and evolving with rules that reflect how cities are governed (Tan & Portugali, 2012).

However, there is still a problem: while these games help people learn and participate, it is not clear if they can produce real, practical design solutions for complicated urban issues where many different interests compete. To fully succeed, serious games need to move beyond just involving people and experimenting they must create real results that influence urban design and planning in meaningful ways.

Conclusion

This study explored the potential of serious games as tools for participatory decision-making and promoting spatial justice. Drawing on a wide range of interdisciplinary research, it became clear that serious games can do more than engage citizens. Indeed, they provide structured, creative platforms for dialogue, negotiation, and collaborative urban design. By turning complex social and spatial issues into interactive experiences, these games empower citizens, especially those from marginalized communities to take an active role in shaping their own societies.

The findings suggest that serious games can help bridge the gap between planners and residents by fostering empathy, shared understanding, and civic learning. In doing so, they support more democratic governance and sustainable urban transformation. However, one major challenge remains: while serious games are effective for participation and learning, they still struggle to deliver concrete, actionable planning outcomes. To increase their real-world impact, future developments should focus on better connecting game insights with formal planning systems and decision-making processes.

Importantly, serious games should not replace traditional planning methods but complement them. They have the unique ability to stimulate civic imagination, strengthen collective agency, and encourage social innovation. Their ongoing evolution and use in different contexts could play a key role in building more inclusive, just, and participatory urban futures.

This perspective aligns with "*Communicative Planning Theory*"[3], which emphasizes inclusive, rational dialogue between citizens and policymakers. The theory holds that all citizens bring valuable local knowledge that, when combined with expert insights, leads to better, more legitimate decisions. It also helps correct power imbalances by amplifying marginalized voices and encouraging civic engagement, social learning, and stronger democratic practices. While it does not replace expert knowledge or advocate for direct democracy, communicative planning supports a collaborative approach that balances professional expertise with public input, an approach that serious games are well-positioned to advance.

The relationship between the concepts is illustrated in the diagram below (Author, 2024).

3 Habermas, J. (1984). The theory of communicative action: Volume 1: Reason and the rationalization of society (T. McCarthy, Trans.). Beacon Press.

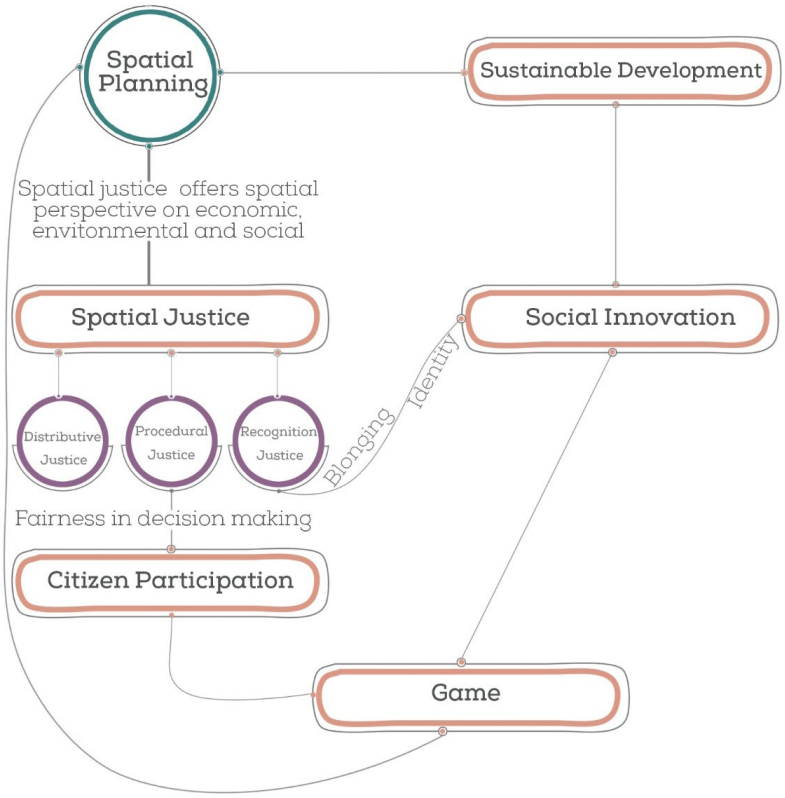


Figure 3: *Literature framework (Author, 2024).*

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